(b) Write a VHDL code to design a BUS structure. 7

(c) Demonstrate the use of guarded block statement in VHDL. 7

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(d) Explain the Delay models in VHDL with suitable example of each. 7

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B. E. (Eighth Semester) Examination, Nov.-Dec., 2015

(Old Scheme)

(Et&T Engg. Branch)

VLSI DESIGN

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Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d). Assume suitable data if required.*

1. (a) What is the basic building block for FPGA? 2

(b) Implement a function

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$$f(x_1, x_2, x_3) = \Sigma\, m(2, 3, 4, 6, 7)$$

using two input LUTs. Assume that the output of

LUT is registered output, draw the required diagram. Also show the boolean manuplation for the implementation. 7

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(c) For the CPLD ; show the programmed section to implement the full adder circuit. 7

(d) Explain the structural domain for designing processor using suitable diagram. 7

2. (a) In N-well process what is depth of N-well on a substrate? 2

(b) Write a short-note on contacts and vi as used in the fabrication process. Also draw the diagram for the same. 7

(c) Define gate-capacitance. Determine the gate capacitance for the following parameters : 7

$t_{ox} = 20$ nm $\quad X_d = 0{\cdot}15\ \mu m$

$L = 1{\cdot}2\ \mu m \quad C_{jo} = 3 \times 10^{-4}\ F/m^2$

$W = 1{\cdot}8\ \mu m \quad C_{jswo} = 8 \times 10^{-10}\ F/m$

$L_D = 3{\cdot}6\ \mu m \quad L_S = 3{\cdot}6\ \mu m$

(d) Explain the procedure of non-rectangular resistance measurement. 7

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3. (a) Differentiate between layout and stick diagram. 2

(b) Draw the schematic of 4 × 4 NAND ROM, also draw the layout for the same. 7

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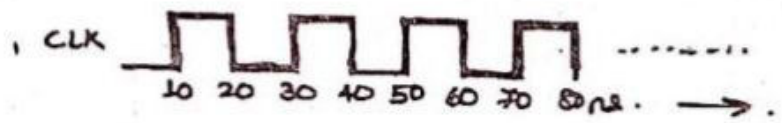
(c) Draw the schematic for 1-bit full adder. Also explain the logic expression. 7

(d) Explain the structure of 6-T SRAM. 7

4. (a) Enlist atleast 4 CAD tools used for VHDL programming. 2

(b) Explain how power is dissipatted in CMOS circuit? 7

(c) Write a VHDL program to generate following waveform. 7



(d) Design a VHDL code for Half-Adder using configuration declaration. 7

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5. (a) Is it mandatory to give label while defining block in VHDL? 2